

R0.74S STEP 16 | TAYLOR 1923 BI-PERIODIC DECAYING VORTEX

FIXED-FRAME BERNOULLI CANCELLATION | VERSION-M MOVING DRIFT | CRITICAL $p = 1$ BOUNDARY

ANALYTIC SCHEMATIC | EXACT IDENTITIES | NOT SIMULATION OR DNS | NO NOVELTY OR PRIORITY CLAIM | NOT CLAY

A Exact vortex
& fixed-frame cancellation

$$W = (\sin x_1 \cos x_2, -\cos x_1 \sin x_2, 0)$$
$$u_A = Ae^{-2(t-t_0)}W$$

**EXACT, SMOOTH, PERIODIC
MEAN-ZERO, UNFORCED**

$$\nabla \cdot (B_W W) = 0$$

**kinetic + physical-pressure
shell fluxes cancel after
periodic integration**

**FIXED COORDINATES
Bernoulli shell flux = 0
PROVED S.417--S.425**

Taylor 1923 bi-periodic
decaying vortex

B Version-M cutoff
leaves moving drift

$$\dot{\xi} = \mu_R b_A W(\xi)$$
$$\mu_R \rightarrow 1 \text{ as } R \downarrow 0$$

MOVING-CUTOFF DRIFT

$$\dot{F}_{k,R} = \frac{\gamma_k \mu_R \eta_R b_A^3}{2R}$$
$$\times W(\xi) \cdot \nabla_{k,R}(\xi)$$



$N + 1$ physical annuli

$C_{k,R} > 0$ on $N + 1$ shells
terminal length δ/A
 $|\dot{F}_{k,R}| \gtrsim RA^3$
PROVED S.429--S.432

Drift is part of the frozen
Version-M observable.

C Scaling split
 $p > 1$ fails; $p = 1$ survives

$$H_{p,N,R}^F \gtrsim A^{3-1/p}$$
$$P_R^M \lesssim RA^3$$

all payment rows included

$$H_{p,N,R}^F / (P_R^M)^{2/3}$$
$$\gtrsim A^{1-1/p} \rightarrow \infty \text{ for every } p > 1$$

S.342 = FALSE

CRITICAL ENDPOINT $p = 1$

$$H_{1,N,R}^F \asymp A^2, \quad P_R^M \asymp A^3$$

**amplitude saturation only
S.444 = OPEN**

S.407, Q.12, Q.1, regularity,
and Clay remain OPEN.

Quantifiers: for every $p > 1$, finite N , and C , choose admissible R , then a sufficiently large amplitude A .

Audit boundary: finite certificates do not machine-prove the continuum payment estimate.